CS602 AY2026/27 August Term

Group Project Guidelines

1. Project Group Formation:
    - 4 to 5 students per group, self-forming

2. Project Topics: **Capacitated Vehicle Routing Problem (CVRP)**
    - Given pairwise distances between all locations, a depot, customer demands, and a set of vehicles with limited capacity, produce a set of routes that minimizes the total travel distance.
    - Each vehicle starts from and returns to the depot.
    - Each customer must be visited exactly once.
    - The total demand served by each vehicle cannot exceed its capacity.

3. Project Output:
    - Code that implements your CVRP algorithm
    - Data generators for CVRP input instances
    - Small instance: expect your code to produce output within 30 seconds
    - Medium instance: expect your code to produce output within 1 minute
    - Large instance: expect your code to produce output within 3 minutes
    - These datasets should be submitted together with the command and the random seed that generates them.
    - It's your job to make sure your code produces output within the expected duration, otherwise, your code will be penalized.

4. Data Specification:
    - Each input instance starts with a list of customer demands, where the first value corresponds to the depot and is always 0. Example:

      ```
      0 2 2 3
      ```

    - This is followed by a symmetric distance matrix, where each entry represents the pairwise distance between two vertices. Vertex 0 is the

depot. Each line is one row, and values are separated by spaces. Example:

```
0 2.5 5.6 6.0
2.5 0 3.9 6.5
5.6 3.9 0 5.2
6.0 6.5 5.2 0
```

- The vehicle capacity and number of vehicles should also be provided. Example:

```
Capacity: 5
Vehicles: 2
```

- Output should contain the optimal total cost as the first line, together with the routes of all vehicles. Each route starts from and returns to vertex 0. Example:

```
21.8
0 1 0
0 2 3 0
```

5. Project Evaluation:
    - The score of a team is calculated based on all instances generated by all groups.
    - For each instance *i*, it will be fed into the code from each team *j* to obtain *n* pairs of scores (*n* is the number of teams), namely $(t_{j,i}\ ,\ c_{j,i})$

$t_{j,i}$	The run-time of team *j* on instance *i*
L_i	The time limit of instance *i*
$c_{j,i}$	The total cost of the feasible CVRP solution produced by team *j*, calculated as the sum of the distances of all vehicle routes
$b_i = \min\{c_{1,i}, \ldots, c_{n,i}\}$	The best feasible solution cost obtained by all teams

    - A submitted solution is considered feasible only if:
        - every customer is visited exactly once;
        - every route starts from and returns to the depot (vertex 0);
        - the total demand served by each vehicle does not exceed the vehicle capacity; and

    - the number of routes does not exceed the given number of vehicles.
- The score of team *j* for this instance *i* is the reciprocal of

$$\frac{1}{2}[2^{\max(0,\frac{t_{j,i}}{L_i}-1)}+\frac{c_{i,j}}{b_i}]$$

- If the output does not represent a feasible CVRP solution, the score for that instance will be 0.
- If the program does not terminate within 1.5 times of the time limit (e.g., 4.5 minutes for a large instance with a 3-minute time limit), the score will be set at ¼. The maximum possible score for each instance is 1.
- The score of each team is averaged separately over all small, medium and large instances. Each team therefore obtains three scores: one for small instances, one for medium instances, and one for large instances.
- The final score is the weighted average of these three scores, with weights of 0.1, 0.3, and 0.6 for small, medium and large instances, respectively.
- The machine specification will be decided at a later stage, most likely a virtual machine on Google Cloud or Amazon AWS.